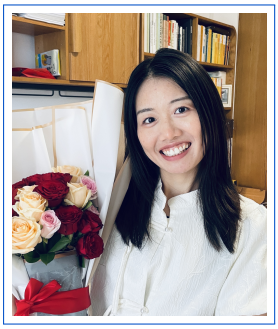
Fang Yang



Curriculum Vitae

Personal Information

Adresse	Vivatsgasse 7, MPIM, Bonn, 53111, Germany
E-mail	f.yang@mpim-bonn.mpg.de
Birthday	May 11, 1996
Homepage	https://fangyang-math.github.io/

Education

2014-2019	**Bachelor of Science**, *Sun Yat-Sen University*
2019-2024	**Doctor of Philosophy**, *Tsinghua University*, Supervisor: Fan Xu
2024-now	**Postdoc**, *Max-Planck Institute for Mathematics*, Bonn

Interests

Algebraic Representation Theory	Calabi-Yau category and tilting theory, coherent sheaves on weighted projective lines, preprojective algebra and Hall algebra.
Geometric Representation Theory	Categorification, semicanonical basis and Nakajima quiver variety, cohomological Hall algebra
Combinatorial Theory	Quantum cluster algebra and Donaldson-Thomas invariants.

Publications

1. **Applications of spherical twist functors to Lie algebras associated to root categories of preprojective algebras**, *with Fan Xu*, Journal of Algebra, Volume 622 (2023): 556-586
2. **Motivic Cluster Multiplication Formulas in 2-Calabi-Yau Categories**, *with Jie Xiao and Fan Xu*, Int. Math. Res. Notices (2024): 13785-807

Preprints

1. **Quantum cluster algebras associated to weighted projective lines**, *with Fan Xu*, arXiv: 2207.02837, submitted
2. **KLR-Schur algebra of coherent sheaves on the projective line: Tilting and PBW bases**, *with Olivier Schiffmann*, arXiv: 2603.01265, submitted
3. **Degenerations in graded quiver varieties and in derived categories of Dynkin quivers**, *with Alessandro Contu*, arXiv: 2602.23939
4. **Formulas in quantum Grothendick ring via derived Hall algebras of type A**, *with Alessandro Contu*, In Preparation

Presentations

Jun. 2023 **Representation Theory in Lyon**, *Université Lyon 1 in Villeurbanne*

Nov. 2023 **The 16th National Algebra Academic Conference**, *Huaqiao University*

Mar. 2024 **Workshop on bricks and endofinite representations**, *Bielefeld University*

Oct. 2025 **Collaborations in Algebra, Representation theory and Ethics**, *ENS Lyon*

Else

Research visits

2023.2-2023.6 **Visiting Ph.D. Student**, *Université Paris-Saclay*, under guidance of Professor Olivier Schiffmann

Conferences

Jul. 2023 **Geometric Representation Theory and Applications**, *Tsinghua University*

Jul. 2023 **International Congress of Basic Science**, *Beijing City government*

Nov. 2023 **Algebraic Representation Theory**, *Beijing Normal University*

Aug. 2024 **International Conference on Representations of Algebras (ICRA 21, 2024)**, *Shanghai Jiao Tong University*

Mar. 2025 **Algebra Days in Caen 2025: Categorization and quantification in cluster theory and beyond**, *University of Caen-Normandy*

Apr. 2025 **Enveloping Algebras and Geometric Representation Theory**, *Mathematisches Forschungsinstitut Oberwolfach*

Jun. 2025 **Enumerative Geometry and Geometric Representation Theory**, *Università di Pisa*

Sep. 2025 **ARTA X: Advances in Representation Theory of Algebras**, *University of Cologne*

Seminars

Spring 2022 **Deriving DG categories**, *Bernhard Keller*,
Reference: Deriving DG categories, Bernhard Keller.

Fall 2022 **Cluster algebras and scattering diagrams**, *Yutong Yu*,
Reference: Wall-Crossing Structures in Cluster Algebras, Lang Mou.

Fall 2022 **Quiver varieties and Coulomb branches**, *Dylan G Allegretti, Peng Shan*,
Reference: K-theoretic Coulomb Branches of quiver Gauge theories and cluster varieties, Gus Schrader, Alexander Shapiro.

Fall 2023 **Integral homology of loop groups and affine Grassmannians**, *Boming Jia, Peng Shan*,
Reference: An introduction to affine Grassmannians and the geometric Satake equivalence, Xinwen Zhu.